Nguyễn Thiện Nhân - 2452879 - W07 drill

Date:

A. ~~Agoetic~~ chromosome is an encoded representation of a candidate solution in a GA. For 5 features, a chromosome like 10110 means select feature 1, 3, 4, and 2 and 5 are excluded. → each chromosome represent a subset of features for model train and eval.

B. Fitness function evaluates how good a candidate solution is and suitability for reproduction. It may combine accuracy with penalty for feature selection. Gradient descet use objective function and gradients to guide parameter update, while GA doesn't need ~~de~~ differential because it use crossover, mutation, selection.

D. Premature convergence is when model loses population diversity and converges to sub-optimal solution. It happens because exploitation is strong compared to exploration → model focus on & similar solutions and overlook better ones. To solve; increase mutation, preserve diverse population, reduce selection pressure

C. Selection choose individuals based on fitness → higher chance of picking better solutions.

TEKO SOLUTION

Date: ____

Crossover: Combines ~~of~~ genetic material from 2 parent chromosomes to ~~eee~~ create offsprings, while mutation randomly changes 1 or more genes to introduce genetic variation. Crossover and mutation create new individual. Selection primarily chooses existing individuals rather than modifying

E. Original 1010. If 2nd brt is flipped → 1110.
Mutation introduces genetic variation by changing a gene → explore new candidate solutions.